Certified Multiarticle Intertwining

Six Proof-Carrying Low-Fan-In Shells Across GBG, MUSIC, SYMPHONY, and TISO

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A four-island extension of *The Multiarticle Theory #beacon91042014201224789*
Research exposition and proof-ledger demonstration — 5 November 2026

See the AI disclaimer that follows this essay.

Abstract

We join four research manuscripts into a typed, proof-bearing Multiarticle: generalized board games and certified self-embedding (GBG), possible dream families (MUSIC), legally executable cooperative scores (SYMPHONY), and tensor-induced allocation (TISO). The construction explicitly retains original theorem vertices, deductive surplus, and the six low-fan-in consequence layers $+1, \ldots, +6$, *with their proof edges and premises*. We explain why taking only a union of theorem statements is inadequate, and why adjoining six layers as mere sets can be redundant if surplus already includes them. An explicit six-bridge, binary-fan-in ladder transports legal traces to dreams, MUSIC, score realizations, event credits, and a static TISO optimizer under stated and increasingly strong compatibility contracts. A separate finite Horn-style counterexample demonstrates that identical MUSIC and identical additive TISO inputs need not imply identical reachable certified SYMPHONY performances. The proof-bearing zipper conserves object-theoretic consequence only when all new deposits remain independently checked against the frozen base. These are mathematical constructions and conditional paper-level proofs, not a claim of a running federation, new settlement of J4, experimental speedup, calibrated Poisson arrivals, or externally machine-checked certificates.

Keywords. multiarticle theory; theorem hypergraphs; low-fan-in consequence mining; deductive surplus; certified proof search; generalized board games; dreaming; symphonies; TISO.

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1 The four islands and the fifth object: their seam

The original Multiarticle program describes six island manuscripts, low-in-fan mining through six levels, sorting by theorem count, recursive theorem-graph zipping, and eventual consequence closure [1]. This article is an explicitly designated *four-island instance*: the four current sources are denoted G (GBG), M (MUSIC), S (SYMPHONY), and T (TISO). Nothing requires pretending that four source articles are six. A six-island instance could later replace this choice without changing the general construction.

Island	Exported source-grounded structure	Critical boundary
G	Labelled legal traces, certified BANK observation, proper reachable self-copy	An embedded root is transported; it need not be the original root [2].
M	D_c of report-union dreams, $\mathrm{MUSIC}(c) = \mathcal{P}(D_c)$, realizability complex K_c	A possible dream collection need not be jointly realizable [3].
S	Legal federation runs, causal score $\mathsf{S}(\rho)$, independently accepted packets	A score is not automatically a proof or a faithful replay [4].
T	Event-indexed settlement tensor, calibrated weights W_j, concave TISO surrogate	Predictions are not one-hot certified outcomes; Poisson calibration is assumed, not established [5].

The *fifth object* is not a fifth manuscript. It is the *typed seam*, a small collection of compatibility statements saying when a move, report, score, certified event, or allocation in one island represents the corresponding object in another. A name shared by two articles is not a proof of such compatibility.

1.1 What is meant by "plus one through plus six"?

A low-fan-in proof step uses at most r earlier claims as explicit parents, with $r = 2$ here. A shell index $k \in \{1, \dots, 6\}$ counts *new bridge applications* beyond the admitted source-theorem base; it does not count every primitive line of a natural-deduction proof. The ladder below supplies a reproducible *upper bound* k on bridge depth for each displayed result, not a proof that no shorter derivation exists in some richer calculus. This distinction prevents an artificial "new theorem" count.

2 A proof-bearing zipper, not a set union

Definition 2.1 (Typed theorem and proof records)**.** *An island is a typed proof hypergraph* $\mathcal{H}_i = (V_i, E_i, \tau_i, \Pi_i)$*. A vertex is a contextual claim*

$$v = (\varphi, \Sigma, \mathcal{L}, V, \mathit{status}, \mathit{origin}),$$

where Σ *is the exact permitted premise scope,* $\mathcal{L}$ *is the language/typing context, and* V *identifies its verification contract. An edge* $e = (P \Rightarrow v, \pi, \kappa)$ *carries a finite parent set* $P \subseteq V_i$*, an explicit proof artifact or paper-level argument* π*, and a declared checking/derivation cost* κ*. A claim may have several distinct proofs.* Identifying equal conclusions must never erase their proof edges.

Definition 2.2 (Six-shell enrichment)**.** *Let* V_i^0 *be the claims initially extracted from source* i*, and let* $\mathfrak{S}_i$ *be a separately recorded deductive-surplus collection. Given a frozen candidate grammar, admissible bridge rules and a proof-checking policy, write* $L_{i,k}^{(2)}$ *for the* proof records *of newly reached claims having a selected derivation of at most* k *new bridge applications, each with at most two parent claims. The*

augmented island is

$$\widehat{\mathcal{H}}_i^{\,6} = \mathrm{ProofUnion}\left(\mathcal{H}_i^0,\, \mathfrak{S}_i,\, \bigcup_{k=1}^{6} L_{i,k}^{(2)}\right). \tag{1}$$

ProofUnion *merges identical* typed claims *without losing distinct proofs, support provenance, costs or original source identity. New source-language formulas may be mined before joining; additional cross-island results may be mined after joining. The latter are recorded separately as* $L^{(2)}_{\mathrm{cross},k}$.

Proposition 2.3 (Extensional absorption, intensional gain)**.** *If* $\mathfrak{S}_i$ *already denotes exactly the union of the six shell* conclusion sets*, then*

$$V_i^0 \cup \mathfrak{S}_i \cup \bigcup_{k=1}^{6} \mathrm{Concl}(L_{i,k}^{(2)}) = V_i^0 \cup \mathfrak{S}_i.$$

Nevertheless the proof-hypergraph union can retain additional distinct proofs and dependency/cost information.

Proof. The displayed equality is idempotence of set union. For the second claim, take a conclusion q with two distinct derivations, one from $p, p \to q$ and another from $r, r \to q$. Both unions have the same conclusion vertex q, while the proof union retains different support sets. If the source premises have different availability or checking costs, the two proofs can have different operational value. Statement-set equality cannot encode this distinction. □

The zipper is then defined, with every identification and cross-edge recorded,

$$\mathcal{Z}^{(6)} = \mathrm{Zip}\big(\,\mathrm{Zip}(\widehat{\mathcal{H}}_G^6, \widehat{\mathcal{H}}_M^6), \mathrm{Zip}(\widehat{\mathcal{H}}_S^6, \widehat{\mathcal{H}}_T^6)\,;\,\mathcal{B}\big)\ \cup_{\mathrm{proof}} \bigcup_{k=1}^{6} L^{(2)}_{\mathrm{cross},k}. \tag{2}$$

Here $\mathcal{B}$ consists of explicit, proven or clearly assumed interface contracts. The four-island pairing makes the two conceptual extremes meet first: formal reachability with dream possibility, and legal performance with resource economics. This is an explanatory pairing, not a theorem-count ranking; the six-paper source ranking procedure can be applied once a stable extraction protocol is supplied.

Proposition 2.4 (Fixed-bridge zip-order invariance)**.** *For a* fixed*, scope-correct collection of equivalences and bridge hyperedges* $\mathcal{B}$*, and a fixed monotone closure operation* Cn*, the consequence set of the total proof union is independent of parenthesization or ordering of the pairwise unions. The recorded heuristic discovery order and finite-budget results need not be invariant.*

Proof. At the claim-and-proof-record level, a tagged disjoint union followed by a fixed equivalence quotient is associative and commutative up to canonical isomorphism. Hence the same premises and inference edges are present after any order of union; application of the same monotone closure has the same least closed superset. A resource-limited miner, in contrast, may stop after different explored prefixes, so equal eventual closure does not imply equal discovered subsets or charged work. □

3 Frozen objects and six compatibility contracts

Fix a set-sized, finitely branching verifier-labelled board G with complete states X, legal labels μ, partial transition maps r_μ, checker observation V_G, BANK view K_G, and charged move costs c_G. The GBG article constructs the address-extension $X = \{0,1\}^* \times Q$ and the proper embedding

$$e(w,q) = (0w, q), \qquad r_\mu \circ e = e \circ r_\mu \quad \text{(where defined)}. \tag{3}$$

It preserves legality, observations and corresponding per-step costs, while $e(s_0)$ is a *reachable transported start*, not a fixed root [2].

For a dreaming condition c, write $\mathcal{E}_c$ for the legal episode set, U_c for typed report atoms, $d_c(h) = \bigcup R_c(h)$ for the union of an episode's reports, and

$$D_c = \{d_c(h) : h \in \mathcal{E}_c\}, \qquad \mathrm{MUSIC}(c) = \mathcal{P}(D_c). \tag{4}$$

The realizability complex K_c consists of finite dream collections co-occurring in some allowed run. A federation F with a complete start c_F has legal runs $\mathrm{Runs}(F, c_F)$, score map S_F retaining designated causal and evidence fields, and

$$\mathrm{SYMPHONY}(F, c_F) = \{S_F(\rho) : \rho \in \mathrm{Runs}(F, c_F)\}. \tag{5}$$

These definitions are from MUSIC and SYMPHONY respectively [3, 4].

The six contracts below are *hypotheses for the ensuing combined theorem*, not properties automatically guaranteed by the four source articles:

C1. e preserves/reflects the finite local legal labels, checker/BANK views and costs as in (3). Its image is forward-closed and the transported start is used.

C2. The map e_* bijects the permitted local episode sets at the two transported conditions; there is a bijection $t : U_c \to U_{c'}$ such that $R_{c'}(e_* h) = \{t[r] : r \in R_c(h)\}$ for each episode h.

C3. A bijection Φ between legal *federation* runs preserves precisely the mapped dream-occurrence sets. (The local GBG map alone does not supply a global run bijection.)

C4. Φ also preserves host event order, causal send/receive relations, authorized packet scopes, checker decisions, all charged global operations, and the score representation up to a declared bijection T_S on realized scores: $S_{F'}(\Phi\rho) = T_S(S_F\rho)$.

C5. A stable-identity bijection transports every accepted logged event together with its weight and attribution shares, and the frozen tensor-calibration operator commutes with the transport. This is an additional *calibration equivariance* requirement.

C6. The allocation channels and feasible sets are transported by a coordinate permutation P, preserving each W_j, efficiency k_j, full cost c_j, minimum m_j, coupled constraints and total budget; the accepted-certificate interpretation stays frozen.

Scope caveat. C1 is furnished by the explicit GBG address-extension construction. C2–C6 require corresponding report encodings, federation rules, score types, identity ledgers and optimizer declarations. Their satisfaction in a particular AMLD+0 system has *not* been demonstrated here.

4 The new intertwined +1 through +6 proof ladder

Each level uses one previous package R_{k-1} and one new contract Ck, so its *bridge-rule in-fan is two.* A package retains all preceding consequences, rather than suppressing hypotheses used several stages earlier. The detailed proofs below are human-readable mathematical arguments, not external proof-assistant certificates.

Theorem 4.1 (+1: transported trace profiles)**.** *Under C1, finite labelled execution traces starting at x and at $e(x)$ are in bijection by action-word identity. Corresponding prefixes have the same verifier/BANK observations and the same cumulative transition charges.*

Proof. For the empty action word the assertion is immediate. Suppose a legal length-n word takes x to y and takes $e(x)$ to $e(y)$. Reflection of enabled labels says precisely the same next labels μ are legal from y and $e(y)$; commutation gives $r_\mu(e(y)) = e(r_\mu(y))$. This proves the trace correspondence by induction and also its converse. At each corresponding vertex observations coincide, while per-step cost preservation and summation yield equal cumulative charged cost. This is a source-derived reconstruction of GBG's exact trace property, now installed as the first cross-island proof node [2]. □

Theorem 4.2 (+2: dream-space transport)**.** *Under C1–C2, the pointwise report image $t_* : D_c \to D_{c'}$ defined by $t_*(d) = t[d]$ is a bijection, and $d_{c'}(e_* h) = t_*(d_c(h))$ for every legal episode h.*

Proof. By C2, image commutes with the union of reports:

$$d_{c'}(e_* h) = \bigcup_{r \in R_c(h)} t[r] = t\left[\bigcup_{r \in R_c(h)} r\right] = t[d_c(h)].$$

Because e_* is onto the corresponding legal episode space, every $d' \in D_{c'}$ has this form. Since t is bijective on atoms, equality $t[d_1] = t[d_2]$ implies $d_1 = d_2$, establishing injectivity. No claim that an untagged dream determines the full episode is needed. □

Theorem 4.3 (+3: MUSIC and compatibility transport)**.** *Under C1–C3, $\mathcal{P}(t_*) : \mathrm{MUSIC}(c) \to \mathrm{MUSIC}(c')$, $A \mapsto \{t_*(d) : d \in A\}$ is a bijection. If the run correspondence in C3 preserves dream co-occurrence, it also transports the realizability complex exactly: $K_{c'} = \{\mathcal{P}(t_*)(A) : A \in K_c\}$.*

Proof. Theorem 4.2 gives a bijection of dream sets, and the direct-image map of a bijection induces a bijection of their powersets (inverse $\mathcal{P}(t_*^{-1})$). For finite $A \subseteq D_c$, $A \in K_c$ means that one allowed run contains all its dreams. C3 carries that run to a run containing precisely the translated dream occurrences, proving forward membership. Apply Φ^{-1} for the reverse inclusion. Without C3, the last sentence is invalid: equal MUSIC need not imply equal K [3]. □

Theorem 4.4 (+4: cooperative score and certificate transport)**.** *Under C1–C4, the score map induces a bijection*

$$T_S : \mathrm{SYMPHONY}(F, c_F) \xrightarrow{\sim} \mathrm{SYMPHONY}(F', c'_F).$$

If $\mathrm{cert}_g(\rho)$ denotes checker-accepted proof status for the same decoded target and scope, then T_S also bijects the existentially defined certified-performance subfamilies.

Proof. Every score on the left is $S_F(\rho)$ for a legal run. C4 gives $T_S(S_F(\rho)) = S_{F'}(\Phi\rho)$, which is a legal score on the right. The run bijection Φ supplies surjectivity; injectivity is part of the declared bijection T_S on realized scores. C4 preserves checker decisions and scoped packets, so $\mathrm{cert}_g(\rho) = \mathrm{cert}_g(\Phi\rho)$. Restricting the score image sets to runs with value one gives the certified-subfamily statement. The condition is stronger than equality of possible MUSIC and rules out unauthorized event rearrangements [4]. □

Theorem 4.5 (+5: conserved tensor-credit transport)**.** *Under C1–C5, suppose distinct verified discovery events are credited by*

$$\theta_{a,b,r,z} = \sum_{\eta \in E} \omega_\eta \, \alpha_{\eta,a,b,r,z}, \quad \alpha_{\eta,a,b,r,z} \geq 0, \quad \sum_{a,b,r,z} \alpha_{\eta,a,b,r,z} \leq 1, \tag{6}$$

with transported event identifiers, weights, roles, frontier labels and shares. Then every transported θ agrees with its source coefficient, and so does the channel weight $W_{(a,r)} = \sum_{b,z} \theta_{a,b,r,z}$. An unchanged calibration rule with equivariant inputs has the same consequence. No forwarded copy creates a new credited event.

Proof. Use the C5 bijection to reindex the sum in (6). Every summand is unchanged, giving equality of each coefficient and hence each finite sum $W_{(a,r)}$. The attribution bound remains true eventwise under the index bijection. A transported packet is assigned the *same* discovery identifier, so it contributes no second term. This is the event-credit convention of TISO, used conditionally with SYMPHONY's evidence ledger; it does not prove the coefficients empirically calibrated [5]. □

Theorem 4.6 (+6: optimizer equivariance and no free copy advantage)**.** *Under C1–C6, define the static TISO surrogate on a nonempty compact feasible region $\mathcal{U}$ by*

$$H(u) = \sum_j W_j(1 - e^{-k_j u_j}), \qquad F(u) = 1 - e^{-H(u)}. \tag{7}$$

Then $H'(Pu) = H(u)$ and $F'(Pu) = F(u)$ for all $u \in \mathcal{U}$, and

$$\arg\max_{u' \in \mathcal{U}'} F'(u') = P\left(\arg\max_{u \in \mathcal{U}} F(u)\right). \tag{8}$$

Moreover, reachable accepted-certificate profiles and the relative costs of corresponding runs coincide. A pure copy/renaming satisfying these contracts yields no strict intrinsic advantage in the static surrogate or corresponding trace cost. If reaching the transported start from the old root costs $K \geq 0$, this setup charge must be added in an end-to-end comparison.

Proof. By theorem 4.5 and C6, $W'_{Pj} = W_j$, $k'_{Pj} = k_j$ and $u'_{Pj} = u_j$ whenever $u' = Pu$. Substitution into (7) and reindexing prove both objective equalities. C6 ensures P bijects feasible sets, so the maximizers correspond exactly, establishing (8), even if the optimizer is nonunique. Theorem 4.4 bijects accepted run profiles, and theorem 4.1 together with C4 preserves their relative charged actions. Finally, a path that first travels from the original root to $e(s_0)$ includes extra setup moves; their nonnegative full costs do not vanish merely because the embedded trajectories are isomorphic. There may still be advantages from genuinely different schedules, premises, algorithms or measured resources, but none is a consequence of a cost-preserving relabeling alone. □

Remark 4.7 (Status of this ladder). The +1 conclusion is explicitly grounded in the GBG source. The further claims are *newly assembled conditional consequences* of source definitions plus C2–C6; they are not six independently established results about an already implemented AMLD system. The depth labels identify this *chosen* two-parent bridge derivation, not logically minimal proof depths nor a claim of novelty against the global mathematics literature.

5 The low-fan-in proof ledger, with an independently reproducible syntax check

Shell	Previous package	Added contract	Newly derived package
+1	GBG strong embedding	C1: trace-view contract	R_1: trace profile
+2	R_1	C2: report commutation	R_2: dream bijection
+3	R_2	C3: co-run correspondence	R_3: MUSIC and K transport
+4	R_3	C4: causal score equivariance	R_4: SYMPHONY transport
+5	R_4	C5: event-credit equivariance	R_5: tensor weights
+6	R_5	C6: feasible/cost equivariance	R_6: TISO and cost comparison

For machine-readable auditing of the *dependency skeleton*, the accompanying `.json` file encodes these six Horn-style inference rules, each with exactly two premises. The accompanying pure-Python checker confirms the rules form the declared +1 through +6 derivation sequence. This check *does not verify* the semantic claims C1–C6 or the paper-level mathematical proofs; those are written out above, where they can be independently inspected.

A proposed complete per-island mining run must additionally specify: (i) exact candidate formula grammar, (ii) permitted source-theorem encodings, (iii) canonical premise scope and proof calculus, (iv) finite search and cost bounds, (v) independent verifier and certificate format, and (vi) deduplication policy. Without these inputs no report of *all* low-fan-in consequences of four open-ended articles is justified. What is supplied here is a six-shell *selected, explicit* cross-island result graph, not an exhaustive extraction.

6 A separate cross-island theorem: possibility cannot schedule proof by itself

The transport ladder establishes what remains equal under very strong equivalence. Its complementary theorem asks what can differ when only the much weaker MUSIC equivalence is known.

Theorem 6.1 (Identical MUSIC and additive TISO data do not fix certifying reachability)**.** *There are two finite legal-run models A, B with equal possible-dream sets and hence equal literal MUSIC, and with equal* declared *per-channel additive TISO input coefficients, for which a target proof is reachable in a legal SYMPHONY run of B but not in any legal run of A. The common global authorized propositional base may remain fixed and sound.*

Proof. Let a, b, g be propositional atoms and fix authorized background assumptions

$$\Sigma_0 = \{a, b, (a \wedge b) \to g\}.$$

A checker accepts g only on receipt of a scoped derivation containing verified access to *both* premise packets a and b and the Horn implication. This is a constraint on legal proof-discovery runs, not a change of mathematical truth: $\Sigma_0 \vdash g$ in both models.

Both models permit a finite episode yielding dream d_a by accessing the packet a, and another yielding d_b by accessing b. In model A a one-use branch token makes these episodes mutually exclusive within a single run; each run ends after accessing one packet, and no checker action can combine them. In model B a legal run can access d_a and then d_b, assemble $a \wedge b$, and apply the implication; the checker accepts g. Other runs can stop after one packet. Thus

$$D_A = D_B = \{d_a, d_b\}, \qquad \mathrm{MUSIC}(A) = \mathrm{MUSIC}(B) = \mathcal{P}(\{d_a, d_b\}),$$

but $\{d_a, d_b\} \notin K_A$ and $\{d_a, d_b\} \in K_B$. Consequently an accepted performance exists only in B.

Finally declare the same nonnegative additive TISO input weights and efficiencies in both models, for example $W_a = W_b = 1$, $k_a = k_b = 1$, with the same static budget and channel costs. Formula (7) then computes identical surrogate values for each equal allocation. This is mathematically permitted because the TISO weights are predictive inputs requiring independent calibration, not verifier-observed terminal outcomes. The identical surrogate therefore cannot distinguish the incompatible-run obstruction from the co-realizable-run case. No assertion is made that these invented equal coefficients are calibrated estimates of the distinct true success probabilities. □

Corollary 6.2 (A necessary seam for operational forecasting)**.** *No universally correct certifying-reachability classifier can depend* only *on literal MUSIC and the numerical static additive TISO inputs in the unrestricted legal-run class of theorem 6.1. Information such as the realizability complex, the causal score, authenticated packet scope or a correctly specified interaction model must also enter.*

Proof. Any classifier depending only on those inputs gives the same answer for A and B by construction, while the correct certifying-reachability answers differ. □

This extends MUSIC's existing same-powerset/different-realizability example [3] to the SYMPHONY certificate gate [4] and the boundary of an additive TISO surrogate [5]. It is a cross-island connection, not a new result in complexity theory or a proof of speedup.

7 Certified theoremhood, meta-theorems, and the independent gate

Theorem 7.1 (Object-level conservativity of an admitted proof zipper)**.** *Fix a common object-level theory Σ_0 and a sound scope-preserving proof checker. Suppose every object-level BANK seed lies in $\mathrm{Cn}(\Sigma_0)$, and every later object-level deposit originating in any island, shell or cross-island zipper edge is checked as a consequence of Σ_0 and previously admitted consequences, with all imported assumptions discharged or retained in the declared sequent. Then for every finite number n of deposits,*

$$B_n^{\mathrm{obj}} \subseteq \mathrm{Cn}(\Sigma_0), \qquad \mathrm{Cn}(\Sigma_0 \cup B_n^{\mathrm{obj}}) = \mathrm{Cn}(\Sigma_0). \tag{9}$$

Proof. Induct on accepted deposit count. The initial inclusion is assumed. At the next deposit the checker accepts only a derivation from the fixed base and earlier accepted conclusions; those conclusions already lie in $\mathrm{Cn}(\Sigma_0)$, so the new conclusion does as well. This proves inclusion. The closure identity follows by extensivity, monotonicity and idempotence of Cn. Neither a dream, a suggested zipper connection, a numeric predicted reward nor a proposed meta-theorem may bypass this check. □

Essential typing. The six-layer theorem claims are about *maps between meta-level structures*. They live in a newly specified mathematical metatheory and do not themselves become unconditional object-level theorems of Σ_0. Adding an unauthorized cross-island premise as an axiom would change the theory, and equation (9) would no longer follow. This maintains the core distinction between expanded search *access* and expanded formal consequence.

TISO's reporting convention must also be kept separate from its objective. Its manuscript models mutually exclusive certified P/R/I/K reports, distinguishes dependence D as a potentially valuable investigative role, and treats U as unresolved evidence status. The familiar six-label AMLD+0 interface P/R/D/I/C/U can be retained as an operational display policy, but this paper does *not* equate TISO's original one-hot terminal classes with all six labels or declare D a certificate of global essential dependence. The Poisson-opportunity interpretation $1-e^{-H(u)}$ is conditional on a separately unverified stochastic assumption [5].

8 A practical recursive miner and the new research target

```
for each source island i in {GBG, MUSIC, SYMPHONY, TISO}:
  extract typed claims, proof references, contexts, costs
  freeze a candidate grammar and permitted low-fan-in rules
  for k in {1,2,3,4,5,6}:
    enumerate candidate proof DAGs with at most 2 parents/step
    retain only checked proofs; label context and shell k
  augment original vertices + surplus + ALL proof-bearing shells
zip islands with explicit bridge obligations and event-ID deduplication
for k in {1,2,3,4,5,6}:  mine new cross-island proof DAGs
apply monotone consequence closure at the declared recursive limit
report accepted proofs separately from heuristics and open conjectures
```

This pseudocode is a specification. The six worked bridge proofs and the small dependency-checking ledger accompanying this paper are a *worked instance*, not a completed run of the general enumeration procedure. In particular, no finite result count is advertised as exhaustive for the source papers.

Conjecture 8.1 (Costed self-referential symphony advantage; open). There exists a frozen family of federated proof-search tasks and a fully charged, verifier-safe operator $\mathcal{L}$ on encoded symphony conditions for which iterating $c_{n+1} = \mathcal{L}(\mathrm{SYMPHONY}(F, c_n))$ gives a well-typed, realizable starting-condition construction and improves measured certified discoveries per total cost over a precisely specified equal-information nonreflective baseline.

No arbitrary condition, fixed point, agent or theorem is privileged merely by designation; a preferred choice must earn its status by an invariant or calibrated measured comparison. Fixed points, gains and formal certifiability remain separate research questions. This formulation relates the open reflective self-embedding / self-referential symphony questions in GBG [2] to MUSIC's realizability and SYMPHONY's verifier gate, while permitting TISO to propose resource allocation without declaring proof by prediction.

9 Claim ledger and limitations

Statement	Epistemic status in this article
Proof-record absorption and fixed-bridge zip order	Proved mathematical observations under declared definitions.
+1 trace transport	Source-grounded proof, reconstructed from GBG.
+2–+6 transport and optimization	Explicit conditional paper-level proofs; C2–C6 are *not* installation facts.
Same MUSIC/TISO inputs, different proof reachability	Finite counterexample proved in this article; deliberately chosen, nonempirical coefficients.
Full four-island six-shell extraction	*Not performed*; a finite candidate grammar, checker and exhaustive search specification are missing.
External ATP/Lean/Coq certificate and live AMLD+0 run	*Not performed or claimed.* The companion Python checks Horn dependency structure only.
J4 settlement, GR–QM bridge, universal speedup	Not claimed.

Acknowledgment of source and method

The Multiarticle method, the low-in-fan +1 through +6 requirement, and the request to include both surplus and proof-bearing mined theorems in the consequence intertwine were supplied by Brian Tenneson. The four source texts were read in their supplied versions, and the present formal synthesis and conditional proofs were prepared with ChatGPT assistance. The central construction is an *expository mathematical joining* of source results and explicit assumptions. No unobserved experiment or independently operating federation is asserted.

References

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[5] B. Tenneson, *Discrete Settlement Tensors and Constrained Optimization in DATA 3.0 / AMLD: The Tensor-Induced Settlement Objective (TISO)*, original note August 2026, repaired expanded manuscript 16 October 2026; supplied `DATA3_AMLD_TISO_Repaired(2).pdf`.

[6] S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004. Background for the concave optimization framework, not a claim of invention here.

AI-assistance, reliability, and verification disclaimer

This manuscript was prepared with AI assistance at Brian Tenneson's direction. Brian Tenneson specified the Multiarticle research objective and the requirement that low-fan-in theorems at depths +1 through +6 be unioned into the consequence intertwine together with their proofs and provenance. ChatGPT (GPT-6) analyzed the supplied and prior-source PDFs, assembled the mathematical framework, wrote conditional propositions, counterexamples and paper-level proofs, generated the LaTeX source, and prepared the companion dependency ledger and local checker.

The four named papers and the original Multiarticle note are the cited sources; borrowed results are attributed. The six-shell transport ladder includes explicit additional contracts C2–C6, which must not be mistaken for verified properties of any running implementation. The formal shell labels count chosen bridge applications in an illustrative two-parent inference skeleton, not mechanically certified minimum proof depth or an exhaustive search of logical consequences. An optional local program can inspect this dependency skeleton but cannot establish the mathematical correctness of the theorem statements or of their assumptions. No claim is made that an external proof assistant, an independent ATP federation, an authenticated verifier, or a calibrated AMLD+0 system accepted these proofs.

Mathematical accuracy, validity, originality, completeness, and applicability are not guaranteed by authorship, AI-generated exposition, internal review, a compilation success or a source citation. Neither a MUSIC set, a SYMPHONY score, a TISO prediction, an open fixed-point conjecture, nor a custom dependency-check log is a substitute for a correctly scoped external proof certificate. No experiment, real-world resource improvement, settlement of J4, or physical bridge between general relativity and quantum mechanics is established by this paper. Users should inspect each typed premise, proof and citation independently before publication or downstream reliance.

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